TEACHING PHILOSOPHY

My teaching centers on project-based learning: students use foundational concepts to pursue questions and build systems they care about. I design assignments that accumulate into a meaningful project, giving each new concept a purpose and students repeated opportunities to apply it. Shared learning goals guide projects shaped by students' interests. I want students to judge whether a method fits a problem and explain what the resulting evidence supports. As students develop these abilities, I gradually give them greater responsibility for directing their work. Robotics offers a particularly effective setting for this approach because students must integrate computational ideas into systems whose physical behavior tests their assumptions.

CLASSROOM TEACHING

As a lead instructor and curriculum developer for AI4ALL, a nonprofit expanding access to AI education, I taught Discover AI, an introductory course combining artificial intelligence and ethics. I wanted students with limited technical backgrounds to develop foundational skills through projects they found meaningful. Because the course was generally noncredit, I also wanted students to see it as a way to pursue their interests. Teams chose project topics, and I helped refine their proposals to meet shared technical goals. Weekly mini-projects built toward a semester-long project that required students to connect their technical work with ethical questions about its application. Students could also use their completed projects as portfolio pieces to explain their capabilities in internship interviews.

After short lectures consolidating material students had studied independently, project teams worked in breakout rooms while the TA and I visited to discuss their progress and understanding. In classes of about 50, this made help accessible to students hesitant to raise questions publicly. In ethics sessions, every group recorded its discussion on a shared digital whiteboard, which we used to bring its ideas into the full-class discussion. Written contributions and small-group discussions gave students ways to participate without having to address the full class while ensuring that every group contributed.

Cumulative projects also gave me opportunities to identify misunderstandings and respond before they affected later work. An early assignment asked teams to select and prepare a publicly available dataset, addressing missing entries and analyzing its quality and potential biases. Students sometimes found data related to their topic that could not support the predictions they proposed. Team discussions often revealed confusion about numerical versus categorical data. I revisited this distinction through the predictions students wanted to make and the models they were considering. This checkpoint let me assess how students applied a foundational concept and help them address misunderstandings before later assignments depended on it.

Later, when some teams' models performed poorly, I encouraged students to investigate their data. Some teams identified dataset biases and developed additional visualizations to explain their findings to classmates, connecting the implications to case studies from the ethics sessions. Students chose how far to extend these visualizations and their ethical analysis, giving them ownership of how they investigated unexpected results. Their presentations demonstrated that they could investigate unexpected results and explain limitations. This reinforced my view that assessment should examine students' reasoning and interpretation of evidence alongside the technical performance of what they build.

I would carry these lessons into my university teaching by assessing individual understanding at selected project milestones through code walkthroughs and questions about how changing an assumption would affect the result. Students' explanations would guide further instruction. Final presentations would require students to explain how components work together and justify their conclusions about the application and, where relevant, its ethical implications. Discussions at the 2026 Computing Research Association Summit informed my thinking about how generative AI could extend this approach. Students could use these tools to build supporting software for more ambitious projects, while these same milestone assessments would require them to demonstrate independently the concepts each assignment is designed to teach.

RESEARCH MENTORING

My approach to research mentoring follows the same principle of increasing student independence. Across my graduate and postdoctoral work, I have mentored more than 20 graduate students and dozens of undergraduates, including serving as the primary day-to-day mentor for two Ph.D. and three master's students through graduation. I also currently serve as the official second supervisor for a Ph.D. student at King's College London. I help students shape projects around their interests and adjust my guidance as they take responsibility for the research.

My mentorship of Ph.D. student Irving Solis illustrates this progression. I initially worked alongside Irving on implementation while helping shape project goals and theory. As he took greater responsibility for implementation, I stepped back from coding and focused our discussions on theoretical arguments and their connections to broader multi-robot planning questions. Our joint insights into when robots need to coordinate also reshaped my research agenda, making the collaboration a source of learning for me. Irving now pursues independent research directions, including some that differ from my own, while we continue to collaborate.

As Irving became more independent, I also helped him develop as a mentor through our joint work with master's student Mike Qin. We discussed how to recognize when Mike needed more support and when he was ready for a greater challenge, adjusting the scope of his work as he developed. We also discussed how to sustain his engagement and help him take responsibility for his own progress. Mike contributed to publications with Irving and first-authored a paper with both of us that extended this direction. Now a postdoc at the University of Illinois Urbana-Champaign, Irving leads large undergraduate robotics competition teams and teaches a robotics course.

Beyond individual mentorship, I have helped create structured opportunities for students to learn research practices. In the Parasol Lab, I helped develop and teach an introductory motion-planning course whose assignments extended a shared research codebase. Students practiced code review and experimental documentation as part of learning to contribute to ongoing research. I would build on this approach in my own lab, providing new researchers with structured instruction and guidance from both me and senior students. I would make research expectations explicit and adjust guidance to each student's starting knowledge, while helping senior students develop as mentors as their responsibilities grow.

CONTRIBUTING TO THE CURRICULUM

At [University], I would welcome the opportunity to teach [TWO OR THREE EXISTING COURSES WITH NUMBERS AND LEVELS], drawing on my research experience. I could also offer a graduate course in motion planning and multi-robot systems that complements [RELEVANT COURSE SEQUENCE OR CURRICULAR NEED]. Students would first develop a planner for one robot and then extend it to coordinate several, using simulation before physical experiments where feasible. Explaining discrepancies between predicted and observed behavior would require them to revisit assumptions and examine how their components interact. Across classroom and research projects, my aim is to help students turn their interests into work they can understand, evaluate, and improve.